

Introduction:

This risk register outlines potential risks that may impact the successful completion of the App Review Rating Prediction System project. Each risk is categorized, assessed based on its likelihood and impact, and linked to an appropriate mitigation strategy. The document will be updated at the start of each phase as the project progresses.

2: Risk Level

Level	Likelihood	Impact
Low	Unlikely to occur during the project	Minor issues that can be easily resolved
Medium	Could occur but not certain	May cause delays or reduce quality
High	Very likely to occur if not addressed	Significant risk to project success

3: Risk Register

ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy
Data Risks					
R-01	Imbalanced dataset (more high-rated reviews than others) may affect model performance.	Data	High	High	Use techniques such as up sampling the Mid and Low class while keeping the High class.
R-02	Many reviews are too short or low-quality to extract useful features for prediction.	Data	High	Medium	Apply review length weighting as discussed in the client meeting. Filter or down-weight very short reviews during preprocessing.
R-03	Difficulty in detecting sarcastic or misleading reviews, which may lead to incorrect predictions.	Data	Medium	Medium	Flag suspicious reviews and document limitations instead of relying fully on detection.
R-04	Low model accuracy may not	Data	Medium	High	Test different experiments and do hyperparameters tuning.

	meet stakeholder expectations (80-90%).				
Technical Risks					
R-05	The baseline Azure ML pipeline may not be reproducible due to missing components or configuration errors.	Technical	Medium	High	Follow instructions carefully. Seek help from the instructor early if any part of the setup is unclear.
R-06	The model may perform poorly on the baseline, making it difficult to demonstrate meaningful improvements in later phases.	Technical	Medium	High	Switch from regression to classification early if needed. Try different preprocessing options and model types in Phase 3.
R-07	The Azure ML compute cluster may run out of credits before the project is completed.	Technical	Medium	High	Use smaller CPU clusters, set minimum nodes to zero, and shut down resources when not in use. Regularly monitor credit usage.
R-08	Pipeline results may not be consistent across different runs, making it difficult to compare experiments.	Technical	Low	Medium	Fix random seeds where possible. Log all run IDs and pipeline settings in the /experiments/ folders on GitHub.
Process Risks					
R-09	A team member may fall behind on assigned tasks,	Process	Medium	Medium	Conduct regular team check-ins (at least once a week) and reassign tasks early if needed,

	which could lead to delays in overall project progress.				instead of waiting until the deadline.
R-10	Since team members are responsible for different parts of the project, there is a risk that project deliverables may become inconsistent.	Process	Low	Medium	Review each other's deliverables before submitting each phase to ensure consistency.